

Adaptive Kernel Selection for Stein Variational Gradient Descent

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Abstract

A central computational objective in Bayesian inference is the construction of accurate and efficient approximations to posterior probability measures. Stein Variational Gradient Descent (SVGD) is a deterministic variational inference procedure that transports an ensemble of particles so that the induced empirical measure approximates a prescribed target distribution. The SVGD update is specified through a reproducing kernel Hilbert space (RKHS), and its convergence behavior and approximation error depend sensitively on the selected kernel. The standard median heuristic provides an inexpensive bandwidth rule, but it has limited expressive capacity and can yield poor approximations, especially in high-dimensional regimes. We propose an adaptive kernel-parameter selection mechanism over an abstract kernel family. Motivated by convergence analyses based on the kernelized Stein discrepancy (KSD), we optimize kernel parameters by maximizing the KSD at the current particle distribution. The resulting algorithm, Adaptive SVGD (Ad-SVG), alternates between an SVGD particle-transport step and a gradient-ascent update of the kernel bandwidths with respect to the KSD objective. We give a simplified mean-field convergence analysis that extends fixed-kernel KSD descent results to the adaptive setting and establishes convergence of the maximal KSD over the kernel class. The numerical experiments indicate that Ad-SVG systematically improves on standard bandwidth heuristics across the considered tasks.

1 Introduction

Stein Variational Gradient Descent (SVGD) [18] is a deterministic particle method for approximate Bayesian inference and is commonly used as an alternative to Markov Chain Monte Carlo (MCMC) sampling. Given an ensemble of particles, SVGD applies transport maps whose velocity fields are obtained from the functional gradient of the Kullback-Leibler (KL) divergence to the target distribution, with the admissible directions constrained by the unit ball of a reproducing kernel Hilbert space (RKHS). A principal technical limitation of SVGD is the dependence of the induced dynamics on the chosen kernel, which substantially affects empirical performance and theoretical convergence properties [8, 28]. Moreover, the empirical particle measure produced by SVGD often exhibits posterior variance underestimation [2]. These issues motivate the view that SVGD deteriorates as the dimension of the state space increases. We address this limitation by introducing a kernel-parameter adaptation rule that updates the kernel during inference through maximization of the kernelized Stein discrepancy (KSD), thereby producing transport directions better matched to complex and high-dimensional target geometries.

1.1 Related work.

Since its introduction [18], SVGD has been deployed as an approximate Bayesian inference method in numerous machine-learning applications [22, 24, 30, 15]. A substantial theoretical literature now

characterizes SVGD at both the distributional and particle levels. Mean-field convergence has been studied in continuous time [23, 8, 5] and discrete time [16, 32], while finite-particle convergence rates have been derived under several regularity and stability assumptions [3, 33]. The dynamics can also be interpreted as gradient flows on spaces of probability measures [19, 8], in analogy with the variational formulation of Jordan, Kinderlehrer, and Otto [14].

The performance of SVGD depends critically on the choice of the kernel function, as it determines the interaction between particles and the overall convergence of the method (see also Figures 1 and 9). Convergence results typically refer to mean-field convergence with respect to KSD, whose relation to weak convergence depends on the selected kernel [11]. The median heuristic [12] yields a simple bandwidth specification, but it is not supported by a general SVGD convergence theory and its performance can deteriorate with increasing dimension [2, 40, 36]. Several subsequent methods modify the SVGD geometry or transport direction to reduce high-dimensional degradation [7, 37, 10, 21]. Other variants replace or augment the RKHS update with neural-network parameterizations of the transport direction [17, 39].

The closest related approach is the mixture-of-kernels framework of Ai et al. [1]. That method defines a KSD associated with a weighted linear combination of base kernels and optimizes the weights by maximizing the resulting multiple-kernel KSD. In contrast, the present work treats kernel selection as continuous parameter optimization over a prescribed kernel family, rather than restricting the parameter space to finite convex combinations of fixed basis kernels.

1.2 Contributions.

This work studies whether the original SVGD transport mechanism, without additional architectural modifications, can produce competitive posterior approximations when its kernel is selected by a principled adaptive rule. By isolating kernel selection from other SVGD extensions, we assess the extent to which the kernel alone controls transport efficiency, variance representation, and robustness in higher-dimensional inference problems.

The main contributions are the following.

- (i) **Adaptive Kernel Selection Method.** We introduce an SVGD variant that updates kernel parameters by maximizing the KSD during inference. Unlike the median heuristic, which specifies a single scalar bandwidth, the proposed rule can optimize multiple continuous kernel parameters and therefore supports anisotropic and iteration-dependent particle interactions.
- (ii) **Theoretical Analysis.** We analyze the method in a discrete-time mean-field formulation and extend fixed-kernel SVGD convergence arguments to the adaptive-kernel setting. In particular, we prove convergence to zero of the supremal KSD over the parameterized kernel class and, under a Stein logarithmic Sobolev inequality, derive iteration-complexity bounds for the mean-field limit.
- (iii) **Empirical Validation.** We evaluate the algorithm on several numerical inference tasks and show that adaptive kernel selection improves over the median heuristic while reducing the variance collapse observed in standard SVGD.

2 Mathematical background

Let $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ be a symmetric positive definite kernel with associated scalar RKHS $\mathcal{H}_0$. We write $\mathcal{H}$ for the d -fold Cartesian product of $\mathcal{H}_0$, equipped with the product inner product $\langle f, g \rangle_{\mathcal{H}} = \sum_{i=1}^d \langle f_i, g_i \rangle_{\mathcal{H}_0}$, and denote the canonical feature map by $\Phi_k(x) = k(\cdot, x) \in \mathcal{H}_0$. The notation $x \cdot y$ refers to the Euclidean inner product, and $\nabla \cdot$ denotes the divergence operator. We denote by $\mathcal{P}(\mathbb{R}^d)$ the set of probability measures on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ and by $\mathcal{P}_p(\mathbb{R}^d)$ the subset with finite p -th moment. For $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$, the Wasserstein p -distance is defined by

$$\mathcal{W}_p(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|_2^p d\gamma(x, y) \right)^{1/p},$$

where $\Gamma(\mu, \nu)$ is the set of all couplings of μ and ν , namely all probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ whose marginals are μ and ν .

2.1 Integral probability metrics and kernelized Stein discrepancy.

Integral probability metrics (IPMs) [26] quantify discrepancy between probability measures by taking the supremum of integration errors over a prescribed function class. For an intractable target distribution, Stein's method [35] supplies test functions whose expectation under the target is zero, thereby eliminating the need to evaluate the normalizing constant. Specifically, for $\mathcal{S}_{\pi} f := \nabla \log \pi \cdot f + \nabla \cdot f$ and an admissible kernel, one has $\int_{\mathbb{R}^d} \mathcal{S}_{\pi} f(x) d\pi(x) = 0$ for all $f \in \mathcal{H}$ [6, 20]. The corresponding *kernelized Stein discrepancy* (KSD) is defined by

$$\text{KSD}(\mu|\pi) := \sup_{f \in B(\mathcal{H})} \left| \int_{\mathbb{R}^d} \mathcal{S}_{\pi} f d\mu \right|, \quad (1)$$

where $B(\mathcal{H})$ is the unit ball of $\mathcal{H}$. The optimizer is $f^* = \frac{\psi}{\|\psi\|_{\mathcal{H}}}$ with $\psi = \int_{\mathbb{R}^d} \mathcal{A}_{\pi}^k d\mu$, where $\mathcal{A}_{\pi}^k(x) = \nabla \log \pi(x) \Phi_k(x) + \nabla \Phi_k(x) \in \mathcal{H}$ and Φ_k is the feature map of k . Consequently, the value of the variational problem is the RKHS norm of ψ , yielding the equivalent representation $\text{KSD}(\mu|\pi) = \|\psi\|_{\mathcal{H}}$.

2.2 Stein Variational Gradient Descent.

Given a target distribution π and an initial reference distribution μ_0 , SVGD constructs a sequence of push-forward measures $\mu_{n+1} := T_{\sharp} \mu_n$, where $T_{\sharp} \cdot$ denotes push-forward under $T(x) = x + \gamma \psi^{\mu_n}(x)$. Here $\psi^{\mu} = \int_{\mathbb{R}^d} \mathcal{A}_{\pi}^k d\mu$ for $\mu \in \mathcal{P}(\mathbb{R}^d)$ is the RKHS steepest-descent direction. This construction follows from Equation (1): the normalized vector field $\frac{\psi^{\mu_n}}{\|\psi^{\mu_n}\|_{\mathcal{H}}}$ minimizes $\frac{d}{d\gamma} \text{KL}(T_{\sharp} \mu_n \| \pi) \Big|_{\gamma=0}$ over the unit ball of $\mathcal{H}$ [cf. 18, Theorem 3.1], where $\text{KL}(\cdot \| \cdot)$ denotes the KL divergence. Thus,

$$\frac{d}{d\gamma} \text{KL} \left((\text{Id} + \gamma \psi^{\mu_n})_{\sharp} \mu_n \parallel \pi \right) \Big|_{\gamma=0} = -\text{KSD}^2(\mu_n|\pi). \quad (2)$$

Applying these transport maps to an initial particle system $\{X_0^i\}_{i=1}^M$ sampled from μ_0 gives the discrete particle recursion

$$X_{n+1}^i = X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k(X_n^i, X_n^j). \quad (3)$$

3 Adaptive kernel selection for SVGD

Because KL convergence of SVGD is difficult to establish without restrictive assumptions, recent analyses often formulate convergence in terms of the kernelized Stein discrepancy (KSD) [16, 33, 32]. KSD convergence alone, however, does not imply weak convergence in general: the corresponding sequence of mean-field measures may fail to be tight, although tightness is necessary for weak convergence of probability measures on Polish spaces. In addition, Equation (1) shows that the RKHS geometry, and hence the kernel, determines whether convergence in KSD is sufficiently strong to imply weak convergence [11].

We therefore formulate the adaptive SVGD variant over a parameterized kernel family $\{k_\theta \mid \theta \in \Theta\}$. For each $\theta \in \Theta$, KSD_θ denotes the associated *kernelized Stein discrepancy*, ψ_θ^μ denotes the optimal SVGD update direction, and Φ_θ denotes the corresponding feature map. This parameterization makes the kernel an explicit optimization variable, so the particle dynamics can incorporate the effect of the kernel on both discrepancy measurement and transport geometry rather than relying on a fixed kernel throughout inference.

The adaptive rule is based on maximizing the KSD between the current empirical particle measure and the target distribution over the kernel family. Formally, the infinitesimal decrease of the KL divergence under an SVGD perturbation is proportional to the squared KSD at the current measure (see Equations (2) and (7)). Consequently, selecting the parameter that maximizes the KSD also selects, within the specified kernel family, the direction with the largest instantaneous KL descent predicted by the SVGD variational characterization. This interpretation is consistent with geometric analyses of SVGD as a gradient flow, where larger KSD values correspond to more favorable convergence behavior when the induced RKHS geometries are compared [28, 8].

Whereas standard SVGD analyzes or controls the KSD associated with a fixed kernel, the proposed method targets the maximal KSD over the selected kernel class. In contrast to the median heuristic of Liu and Wang [18], the procedure applies to general parameterized kernels: at iteration n , the idealized parameter selection is

$$\theta_n \in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_\theta(\mu_n | \pi).$$

Operationally, before each particle-transport step we apply one or more gradient-ascent iterations with step size $s > 0$ to the KSD objective in the kernel parameter. To reduce computational cost, the parameter update may be skipped on selected SVGD iterations; this choice is represented by the user-specified decision variable `paramupdate` in Algorithm 1. The particle-transport component can be replaced by alternative SVGD implementations, such as adaptive step-size schedules, line searches, momentum, or second-order/preconditioned updates, without changing the kernel-selection mechanism. Likewise, the gradient-ascent optimizer for the kernel parameter can be replaced by another suitable parameter-optimization routine. The implementation uses the explicit KSD representation [6, 20]

$$\text{KSD}^2(\mu | \pi) = \int_{\mathbb{R}^d \times \mathbb{R}^d} u_\pi^k \, d(\mu \otimes \mu) \quad (4)$$

with

$$\begin{aligned} u_\pi^k(x, y) = & k(x, y) \nabla \log \pi(x) \cdot \nabla \log \pi(y) + \nabla \log \pi(y) \cdot \nabla_x k(x, y) \\ & + \nabla \log \pi(x) \cdot \nabla_y k(x, y) + \text{trace}(\nabla_x \nabla_y k(x, y)), \end{aligned} \quad (5)$$

which permits direct differentiation with respect to the kernel parameter. The additional computational cost of Ad-SVGd relative to standard SVGD has two relevant features:

- The kernel-parameter gradient ascent reuses the score evaluations $\nabla \log \pi(X_n^i)$ computed for the corresponding SVGD step, so Ad-SVGd does not require additional evaluations of $\nabla \log \pi$.
- The cost can be controlled by using a small value of `nstepstheta`, by updating the parameter only after several SVGD steps, or, for large particle counts M , by estimating the KSD from a subsample of particles.

Algorithm 1: Ad-SVGd

Input: Initial particle set $\{X_0^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$, kernel class $\{k_\theta \mid \theta \in \Theta\}$, initial kernel parameter $\theta_{-1} \in \Theta$, step sizes $\gamma, s > 0$, number of steps `nsteps`, `nstepstheta` $\in \mathbb{N}$

Output: Final particle set $\{X_{\text{nsteps}}^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$

```

for  $n = 0$  to nsteps  $- 1$  do
    Decide paramupdate  $\in \{\text{True}, \text{False}\}$ ;
    if paramupdate then
         $\theta_n^0 \leftarrow \theta_{n-1}$ ;
        for  $\ell = 0$  to nstepstheta  $- 1$  do
             $\theta_n^{\ell+1} \leftarrow \theta_n^\ell + s \nabla_{\theta_n^\ell} \text{KSD}_{\theta_n^\ell}^2 \left( \frac{1}{M} \sum_{i=1}^M \delta_{X_n^i} \mid \pi \right)$ ;
        end
         $\theta_n \leftarrow \theta_n^{\text{nstepstheta}}$ ;
    else
         $\theta_n \leftarrow \theta_{n-1}$ ;
    end
    for  $i = 1$  to  $M$  do
         $X_{n+1}^i \leftarrow X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k_{\theta_n}(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k_{\theta_n}(X_n^i, X_n^j)$ ;
    end
end

```

4 Convergence analysis

We consider a target measure with Lebesgue density $\pi(x) \propto \exp(-V)$ for a potential $V : \mathbb{R}^d \rightarrow \mathbb{R}$. The analysis extends recent KSD-based convergence results for SVGD to the adaptive-kernel setting. More precisely, we adapt the discrete-time argument of [32], which assumes Talagrand’s inequality for the target; this condition is weaker than the logarithmic Sobolev inequality commonly used in related analyses. The assumptions from [32] are imposed uniformly over the kernel-parameter space.

Assumption 1. *We assume that $V \in C^2$ such that $\int_{\mathbb{R}^d} \exp(-V(x)) \, dx < \infty$ and that the Hessian H of V is uniformly bounded w.r.t. the operator norm, i.e. there exists $L \geq 0$ such that $\|H(x)\|_{\text{op}} \leq L$ for all $x \in \mathbb{R}^d$.*

Assumption 2. *We assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies Talagrand’s inequality T1, which means that there exists $\lambda > 0$ such that*

$$\mathcal{W}_1(\mu, \pi) \leq \sqrt{2 \text{KL}(\mu \parallel \pi) / \lambda}$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Assumption 3. We assume there exists $B > 0$ such that $\|\Phi_\theta(x)\|_{\mathcal{H}_0} \leq B$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$, and that $\nabla \Phi_\theta$ is continuous with $\|\nabla \Phi_\theta(x)\|_{\mathcal{H}}^2 = \sum_{i=1}^d \|\partial_i \Phi_\theta(x)\|_{\mathcal{H}_0}^2 \leq B^2$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$.

4.1 Convergence analysis of SVGD under Talagrand's inequality

We denote

$$\mathcal{F}(\mu) := \text{KL}(\mu \| \pi).$$

and make use of the following fundamental inequality. Given a fixed kernel parameter $\theta \in \Theta$ such that Assumption 3 is satisfied, define the pushforward measure

$$\tilde{\mu} = (I + \gamma g) \# \mu$$

for arbitrary $g \in \mathcal{H}$. Under Assumptions 1 and 3, let $\gamma, B > 0$, $\alpha > 1$ and $g \in \mathcal{H}_\theta$ such that $\gamma \|g\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$. Then, according to Proposition 3.1 in [32], it holds

$$\mathcal{F}(\tilde{\mu}) \leq \mathcal{F}(\mu) + \gamma \langle \psi_\theta^\mu, g \rangle_{\mathcal{H}_\theta} + \frac{\gamma^2 K}{2} \|g\|_{\mathcal{H}_\theta}^2 \quad (6)$$

with $K = (\alpha^2 + L)B$. For the iterates (3) with a fixed kernel parameter $\theta \in \Theta$ one can then derive the descent condition [32, Theorem 3.2],

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_\theta^2(\mu_n | \pi), \quad (7)$$

provided that

$$\gamma \leq (\alpha - 1) \left(\alpha B (1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}}) \right)^{-1}. \quad (8)$$

The key aspect to verify (7) is the verification of $\gamma \|\psi_\theta^{\mu_n}\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$ using Talagrand's inequality, Assumption 2, which allows to apply (6) for $g = \psi_\theta^{\mu_n}$. Condition (7) can then be used to argue that

$$\lim_{n \rightarrow \infty} \text{KSD}_\theta^2(\mu_n | \pi) = 0,$$

since $\sum_{n=0}^\infty \text{KSD}_\theta^2(\mu_n | \pi) \leq c_\gamma^{-1} \mathcal{F}(\mu_0)$ for $c_\gamma = \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) > 0$.

When introducing an adaptive kernel parameter choice (θ_n) , the inequality (7) changes to

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_{\theta_n}^2(\mu_n | \pi).$$

4.2 Convergence analysis of Ad-SVGd.

Suppose the adaptive SVGD iteration can be written in the following simplified form

$$\begin{aligned} \mu_{n+1} &= (I + \gamma \psi_{\theta_n}^{\mu_n}) \# \mu_n, \\ \theta_n &\in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi), \end{aligned} \quad (9)$$

meaning that

$$\|\psi_{\theta_n}^{\mu_n}\|_{\mathcal{H}_{\theta_n}} = \max_{\theta \in \Theta} \text{KSD}_\theta(\mu_n | \pi).$$

We emphasize that this formulation is possible when the maximization of the KSD with respect to the kernel parameter has a (unique) closed-form solution. This is actually the case in the multiple-kernel SVGD framework [1], which parameterizes the kernel as a convex combination of base kernels

$$k_\theta(x, y) = \sum_{i=1}^N \theta_i k_i(x, y) \quad \text{with} \quad \sum_{i=1}^N \theta_i = 1.$$

For the general setting, the kernel parameter update θ_n in (9) needs to be approximated. Note that in the general setup we assume that $\operatorname{argmax}_{\theta \in \Theta} \operatorname{KSD}_\theta^2(\mu_n | \pi) \neq \emptyset$ for all $n \in \mathbb{N}$.

The following descent condition is a direct consequence of Theorem 3.2 in [32]:

Lemma 1. *Suppose that Assumptions 1-3 are satisfied. For any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

there exists $c_\gamma > 0$ such that for all $n \in \mathbb{N}$

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \operatorname{KSD}_\theta^2(\mu_n | \pi),$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (9).

Corollary 2. *Under the same assumptions as in Lemma 1 it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \operatorname{KSD}_\theta^2(\mu_n | \pi) = 0$.*

The situation changes when we assume that we can only approximately solve the maximization task of the KSD with respect to the kernel parameter. Suppose that $\max_{\theta \in \Theta} \operatorname{KSD}_\theta(\mu | \pi) < \infty$ for any $\mu \in \mathcal{P}(\mathbb{R}^d)$ and consider the alternating scheme

$$\begin{aligned} \theta_n &= \Psi_n(\theta_{n-1}, \mu_n), \\ \mu_{n+1} &= (\operatorname{Id} + \gamma \psi_{\theta_n}^{\mu_n})_\# \mu_n \end{aligned} \tag{10}$$

for some sequence of iterative update rules $\Psi_n : \Theta \times \mathcal{P}(\mathbb{R}^d) \rightarrow \Theta$ with the goal to maximize $\operatorname{KSD}_\theta(\mu_n | \pi)$. Specifically, in Algorithm 1 Ψ_n corresponds to the gradient ascent update for the KSD. However, as mentioned above, this update could be replaced by other suitable iterative optimization schemes. Our required assumption on the update rule is the following convergence behavior.

Assumption 4. *We assume that there exists a sequence $(\varepsilon_n)_{n \in \mathbb{N}}$ such that $\lim_{n \rightarrow \infty} \varepsilon_n = 0$ and*

$$\max_{\theta \in \Theta} \operatorname{KSD}_\theta^2(\mu_n | \pi) - \operatorname{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \varepsilon_n \quad \text{for all } n \in \mathbb{N}.$$

Using this assumption, we can make the following convergence guarantee.

Theorem 3. *Suppose that Assumptions 1-3 are satisfied. Under Assumption 4, for any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \operatorname{KSD}_\theta(\mu_n | \pi) = 0$, where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10).

Proof. Using Theorem 3.2 in [32] we obtain

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_\gamma \text{KSD}_{\theta_n}^2(\mu_n|\pi)$$

Since $(\mathcal{F}(\mu_n))_{n \in \mathbb{N}}$ is decreasing and bounded below by zero, it converges and, in particular, it is a Cauchy sequence. Hence, we have

$$c_\gamma \text{KSD}_{\theta_n}^2(\mu_n|\pi) \leq \mathcal{F}(\mu_n) - \mathcal{F}(\mu_{n+1}) \rightarrow 0$$

as $n \rightarrow \infty$. Under Assumption 4 we obtain

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n|\pi) \leq \text{KSD}_{\theta_n}^2(\mu_n|\pi) + \varepsilon_n \rightarrow 0$$

as $n \rightarrow \infty$. □

Under a Stein logarithmic Sobolev inequality, the Ad-SVGD convergence statement can be strengthened to convergence in KL divergence. For specified regimes of ε_n , we derive iteration-complexity bounds for the mean-field limiting system (10). This setting also yields an explicit bias bound when the KSD maximization is solved only to fixed $\bar{\varepsilon}$ -accuracy at each iteration. The refined analysis is given in Section B. We emphasize that the condition only requires one kernel in the class $\{k_\theta\}$ to satisfy the Stein logarithmic Sobolev inequality, which makes the assumption more flexible than a uniform requirement over the entire class.

5 Numerical Experiments

This section evaluates Ad-SVGD on two primary numerical test problems: a one-dimensional Gaussian mixture model and a linear inverse problem induced by an ordinary differential equation. Additional controlled experiments on dimensional scaling are reported in Section A. In Section 5.4, we further compare Ad-SVGD with median-heuristic SVGD on the Bayesian logistic regression benchmark used in prior SVGD studies.

5.1 Kernel parameterization

SVGD is typically implemented with radial kernels of the form

$$k_h(x, y) = \exp\left(-\frac{\|x - y\|_p^p}{h}\right),$$

where $\|\cdot\|_p$ is the p -norm on $\mathbb{R}^d$ [e.g. 18, 2, 8]. The relevant kernel parameter is the bandwidth h . The median heuristic sets $h = \frac{\text{med}^p}{\log(M-1)}$, with med equal to the current median pairwise particle distance; this rule is intended to balance the attraction and repulsion terms in the SVGD update (3) [18].

To use the additional degrees of freedom provided by adaptive selection, the experiments employ product kernels

$$k_h(x, y) = \prod_{i=1}^d \exp\left(-\frac{|x_i - y_i|^p}{h_i}\right)$$

with parameter vector $h = (h_1, \dots, h_d)$, so each coordinate has its own bandwidth. The derivatives required by Algorithm 1 for these kernels, as obtained from Equations (4) and (5), are available in closed form. We also tested a coordinatewise median-heuristic variant for this product kernel, but that rule produced poor approximations and exhibited variance collapse.

The same adaptation mechanism can be applied to more scalable kernel parameterizations in higher-dimensional settings. For example, spectral or low-rank kernels could parameterize dominant variability directions without the full cost of product kernels, while Matérn and mixed-product constructions could introduce explicit smoothness and anisotropy controls.

5.2 Toy example

We first study the one-dimensional benchmark of Liu and Wang [18], namely the two-component Gaussian mixture $\pi = \frac{1}{3}\mathcal{N}(-2, 1) + \frac{2}{3}\mathcal{N}(2, 1)$. The initial particle system is sampled from $\mathcal{N}(0, 1)$, with $M \in \{50, 200, 500\}$ particles. SVGD is run for 10^4 iterations with step size 1, using the kernel family described above with $p = 1$ and fixed bandwidth values h . Approximation quality is measured by the Wasserstein 1-distance $\mathcal{W}_1$, computed using the one-dimensional identity $\mathcal{W}_1(\mu, \nu) = \int_{\mathbb{R}} |F_\mu(x) - F_\nu(x)| dx$ (see Panaretos and Zemel [29]) against an exact sample of size 10^5 . Figure 1 shows that accurate approximation occurs only over a limited bandwidth range. Thus, even in this low-dimensional benchmark, the performance of SVGD is strongly controlled by the value of h , making the bandwidth-selection procedure a critical component of the method.

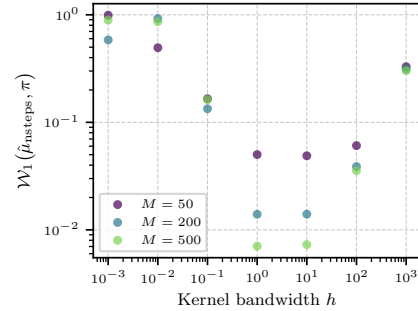


Figure 1: Terminal Wasserstein 1-distances for the one-dimensional example under SVGD with fixed bandwidth values h .

5.3 Linear inverse problem based on ODE

The next experiment is adapted from [38, Example 2.1]. The forward model is defined by the one-dimensional boundary-value problem

$$\begin{cases} -f''(s) + f(s) = u(s) & \text{for } s \in (0, 1) \\ f(s) = 0 & \text{for } s \in \{0, 1\} \end{cases} \quad (11)$$

The associated inverse problem is to infer the forcing term $u(\cdot) \in L^2([0, 1])$ from finitely many noisy point evaluations of the solution $f \in H^2([0, 1]) \cap H_0^1([0, 1])$. The observation model is

$$y = \Phi(u) + \varepsilon \in \mathbb{R}^{N_{\text{obs}}},$$

where $\varepsilon \in \mathbb{R}^{N_{\text{obs}}}$ denotes observational noise. The forward operator $\Phi : L^2([0, 1]) \rightarrow \mathbb{R}^{N_{\text{obs}}}$ is $\mathcal{O} \circ H^{-1}$, where $H(f) = -f'' + f$ and $\mathcal{O}(f) = (f(s_1), \dots, f(s_{N_{\text{obs}}}))^\top \in \mathbb{R}^{N_{\text{obs}}}$ evaluates f at the equidistant points $s_k = \frac{k}{N_{\text{obs}}}$, $k = 1, \dots, N_{\text{obs}}$.

For the Bayesian inverse problem, the unknown forcing term u is assigned a Gaussian-process (GP) prior represented through the truncated Karhunen-Loève (KL) expansion

$$u(\cdot, x) = Ax = \sum_{k=1}^{N_x} x_k \psi_k,$$

where $\psi_k(s) = \sqrt{2} \sin(\pi k s)$ and the coefficients are independent with $x_k \sim \mathcal{N}(0, \lambda_k)$ and $\lambda_k = 50k^{-2}$. The inference variable is the coefficient vector $x = (x_1, \dots, x_{N_x})^\top \in \mathbb{R}^{N_x}$, whose prior is $\mathcal{N}(0, \Gamma_0)$ with $\Gamma_0 = \text{diag}(\lambda_1, \dots, \lambda_{N_x})$. Under Gaussian observational noise $\varepsilon \sim \mathcal{N}(0, \Gamma)$, the posterior density is

$$\pi(x) \propto \exp \left(-\frac{1}{2} \|\Gamma^{-1/2}(y - \Phi Ax)\|^2 - \frac{1}{2} \|\Gamma_0^{-1/2} x\|^2 \right), \quad x \in \mathbb{R}^{N_x}.$$

In the implementation, H is replaced by a numerical discretization of Equation (11) on a grid with mesh size 2^{-8} , and the system is fully observed, so $N_{\text{obs}} = 2^8$. The KL expansion of u uses $N_x = 16$ terms, and the observational noise covariance is $\Gamma = 10^{-3} \text{Id}_{N_{\text{obs}}}$. Reference observations $\bar{y} \in \mathbb{R}^{N_{\text{obs}}}$ are generated by drawing $\bar{x} \sim \mathcal{N}(0, \Gamma_0)$ and setting $\bar{y} = \Phi A \bar{x}$.

We compare median-heuristic SVGD, denoted Med-SVG, with Ad-SVG for particle ensemble sizes $M \in \{50, 100, 200\}$. Both methods use the kernels specified in Section 5.1 with $p = 1$. Each algorithm is run for $4 \cdot 10^5$ iterations with particle-update step size 10^{-3} and the AdaGrad-type adaptive step-size control suggested by Liu and Wang [18]. For Ad-SVG, the bandwidth-update step size is 10^{-5} , and the bandwidth vector is updated once per 100 particle updates. Under this update schedule, Med-SVG and Ad-SVG have no significant runtime difference.

Figure 2 reports the reconstructed GP for a representative random seed. Both methods approximate the posterior mean, but only Ad-SVG reproduces the posterior uncertainty at the displayed scale. To quantify approximation error, we compute the Wasserstein 2-distance $\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)$ between the exact posterior π and the Gaussian distribution $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$ fitted to the particle ensemble, where $\hat{\mu}$ is the particle sample mean and $\hat{\Sigma}$ is the particle sample covariance [29, Section 3.2]. Because π is multivariate normal, this distance has the explicit expression

$$\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)^2 = \|\hat{\mu} - \mu_\pi\|^2 + \text{trace} \left(\hat{\Sigma} + \Sigma_\pi - 2(\hat{\Sigma}^{1/2} \Sigma_\pi \hat{\Sigma}^{1/2})^{1/2} \right),$$

where μ_π and Σ_π denote the posterior mean and covariance. We also compare the marginal variances of the terminal particle distribution with the corresponding posterior variances. Figure 3 aggregates the results over 56 random seeds; the posterior covariance is independent of $\bar{y}$. The results show that Ad-SVG yields a more accurate posterior approximation than Med-SVG, whose terminal particles underestimate uncertainty. Moreover, unlike Med-SVG, Ad-SVG improves when the particle count increases beyond 50. Figure 4 displays the Ad-SVG bandwidth trajectories. The coordinatewise bandwidths stabilize earlier than the approximation error, and their final values vary substantially across coordinates, with a negative association between bandwidth magnitude and the corresponding marginal posterior variance.

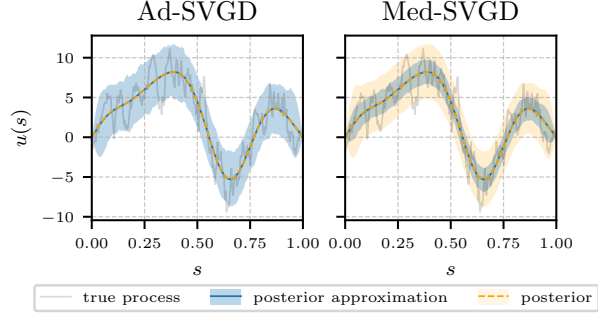
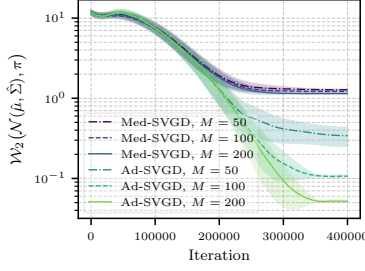
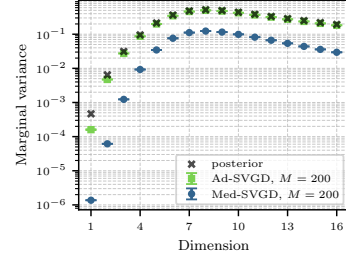


Figure 2: Gaussian-process reconstruction for the ODE inverse problem, with posterior mean and 90% confidence interval.

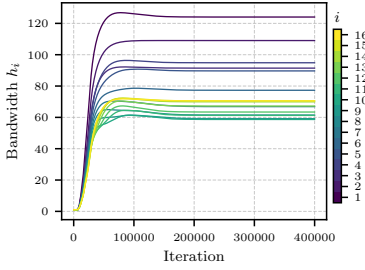


(a) Convergence profile of the approximation error.

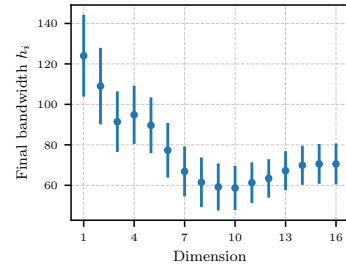


(b) Coordinatewise marginal variances.

Figure 3: Aggregate ODE inverse-problem results for Med-SVGD and Ad-SVGD, reported as mean and 95% confidence interval over 56 random seeds.



(a) Mean coordinatewise bandwidth trajectory.



(b) Final bandwidths, with mean and 95% confidence interval.

Figure 4: Bandwidth-parameter behavior for Ad-SVGD on the ODE inverse problem, aggregated over 56 random seeds.

5.4 Bayesian logistic regression

Following Liu and Wang [18] and Liu et al. [21], we evaluate a Bayesian logistic regression (BLR) model on the Covertypes data set [4]. The regression weights w have conditional prior $w \mid \alpha \sim \mathcal{N}(0, \alpha^{-1})$, with hyperprior $\alpha \sim \text{Gamma}(1, 0.01)$, and the inference target is the posterior distribution of $x = [w, \alpha]$. As in Liu et al. [21], reference posterior samples are generated with the No-U-Turn Sampler (NUTS) of Hoffman and Gelman [13], and these samples are used to assess the approximation quality of Med-SVGD and Ad-SVGD. Following the evaluation protocol of Liu et al. [21], each method is run on random subsets of 1000 observations, and results are aggregated over 10 independent draws. Classification performance is evaluated by test-set prediction accuracy using the particle mean. Figure 5 shows that Ad-SVGD outperforms Med-SVGD for almost all seeds, producing higher test accuracy and a particle distribution closer to the MCMC reference. The same figure reports squared maximum mean discrepancy (MMD²) [25, Section 3.5] to the MCMC reference samples as a posterior-approximation metric. The discrepancy values indicate a substantial difference in how the two particle systems

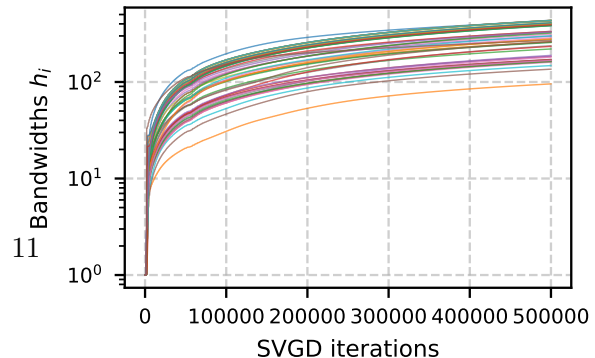


Figure 6: Bandwidth evolution for the BLR experi-

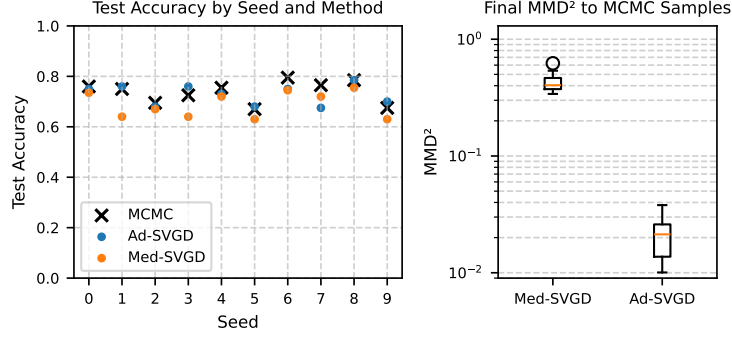


Figure 5: Bayesian logistic regression approximation quality, measured by prediction accuracy (left) and MMD^2 to NUTS reference samples (right).

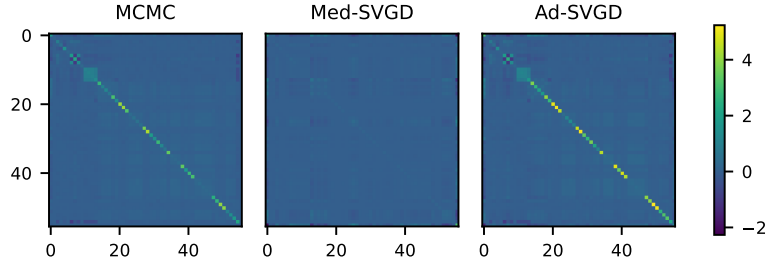


Figure 7: Covariance matrices of terminal SVGD particle systems compared with MCMC reference samples, averaged over 10 seeds.

represent the posterior distribution. This conclusion is supported by Figure 7, which compares final-particle covariance matrices with the MCMC reference after averaging over the 10 seeds, using the visualization convention of Liu et al. [21]. Med-SVGD strongly underestimates posterior uncertainty, whereas Ad-SVGD provides a more accurate covariance approximation.

Finally, Figure 6 shows the coordinatewise bandwidth trajectories: starting from the uninformed initialization with value 1 in every component, Ad-SVGD identifies bandwidths that differ from the initialization by several orders of magnitude.

6 Limitations and outlook

The principal limitation of the theoretical analysis is its dependence on Assumption 4. Ongoing work studies conditions under which Assumption 4 is satisfied by the alternating gradient-ascent procedure in Algorithm 1, which is used in the experiments. Although the current analysis does not prove Assumption 4 for the implemented optimizer, the numerical results indicate that the procedure is effective in the tested regimes.

The present analysis treats the original SVGD dynamics. A natural extension is to combine the proposed kernel-parameter adaptation with more recent transport geometries and update mechanisms, including sliced SVGD [10], Grassmann SVGD [21], and Stein transport [27].

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A More numerical experiments

A.1 Gaussian mixture models

We return to the one-dimensional mixture model from Section 5.2. For the fixed-bandwidth setup, Figure 9 compares terminal particle histograms with the target density for $h \in \{0.001, 1, 1000\}$. We then compare Med-SVGD and Ad-SVGD as the particle count M varies, using the terminal particle distributions after 10^4 iterations with step size 1.

Figure 8 reports the Wasserstein 1-distance between the terminal empirical particle measure and the target distribution for Med-SVGD and

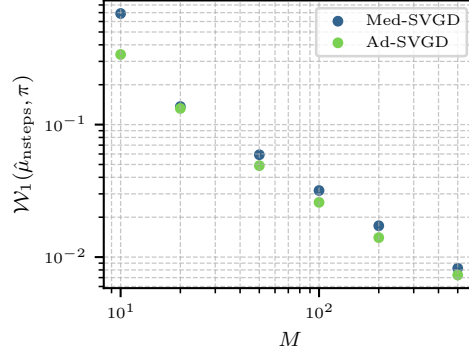


Figure 8: Approximation error for the one-dimensional Gaussian mixture.

Ad-SVGD with $M = 10, 20, 50, 100, 200, 500$. As expected, the approximation error decreases as the particle count increases for both algorithms. The two methods perform similarly in this one-dimensional setting and both reach Wasserstein distance below 0.01 when $N = 500$.

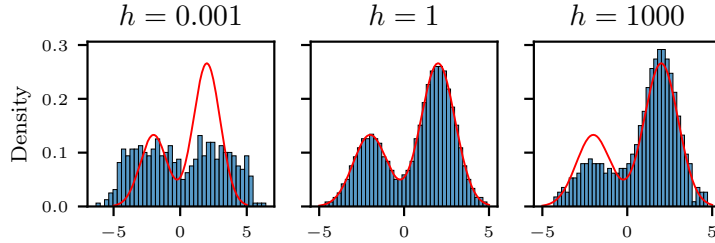


Figure 9: Terminal particle histograms for fixed bandwidths $h = 0.001, 1, 1000$, with target density π shown in red.

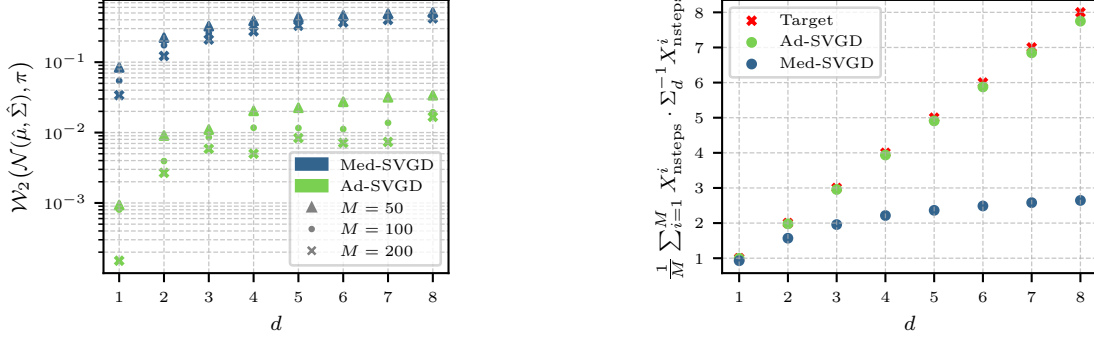
In this one-dimensional experiment, both procedures approximate the target distribution accurately. The adaptive bandwidth rule remains stable, but it does not provide a substantial advantage over the median heuristic in this low-dimensional regime.

A.2 Scaling dimension: multivariate normal distribution

For the dimensional-scaling experiment, the target distributions are Gaussian measures $\pi_d = \mathcal{N}(0, \Sigma_d)$ with $\Sigma_d = \text{diag}(1, \frac{1}{2}, \dots, \frac{1}{d})$ for $d \in \{2, \dots, 8\}$. The initial particle distribution is $\mathcal{N}(0, \frac{1}{d})^{\otimes d}$. Unless numerical stability required a smaller step size and a corresponding adjustment of the iteration count, each algorithm was run for 10^4 iterations with step size 0.1.

We evaluate the terminal particle systems produced by Med-SVGD and Ad-SVGD as d increases. Since the Wasserstein 2-distance between Gaussian measures has a closed form (see [29]), the approximation metric is the Wasserstein 2-distance between π_d and $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, where $\hat{\mu}$ and $\hat{\Sigma}$

are the empirical mean and covariance of the particle system [29, Section 3.2]. Following the terminology of the Python Optimal Transport implementation used in the computations [9], we refer to this metric as the *Bures Wasserstein distance*. Figure 10a reports this metric for Med-SVGD and Ad-SVGD with $M = 50, 100, 200$ as dimension increases; Ad-SVGD yields substantially smaller error for every tested value of d .



(a) Approximation error for $M \in \{50, 100, 200\}$.

(b) χ^2 -test statistic for $M = 200$.

Figure 10: Results for multivariate normal target distributions as the ambient dimension increases.

The Bures Wasserstein distance is relatively insensitive to errors in the low-variance coordinates of π_d . To evaluate uncertainty representation in those coordinates, we compute the statistic $X \cdot \Sigma_d^{-1} X$, which is χ^2 -distributed with d degrees of freedom and expectation d when $X \sim \mathcal{N}(0, \Sigma_d)$ (cf. [34, p. 27]). For $M = 200$, Figure 10b compares the particle average of this statistic with its target expectation. As d increases beyond the one-dimensional case, the Med-SVGD statistic increasingly deviates from d , whereas the Ad-SVGD statistic remains comparatively close to the target expectation. This indicates that Med-SVGD fails to represent uncertainty in the higher-dimensional targets, while Ad-SVGD remains accurate on these examples.

Additional coordinatewise information for the terminal particle systems with $M = 200$ is given in Tables 1 and 2. For each d , the tables report particle marginal variances against the target marginal variances listed in the first row. Table 1 contains Med-SVGD results, and Table 2 contains Ad-SVGD results. Ad-SVGD approximates the target marginal variances in all coordinates, whereas Med-SVGD substantially underestimates target uncertainty, consistent with Figure 10b.

Finally, we inspect the marginal particle distributions for $M = 200$ and $d = 8$. For visualization, each coordinate is rescaled by the inverse of the corresponding target marginal standard deviation, equivalently multiplying the i -th coordinate by i , so that each target marginal becomes standard normal. Figures 11 and 12 display the normalized marginal histograms for Med-SVGD and Ad-SVGD, respectively, with the standard Gaussian density included for reference. Ad-SVGD accurately represents all normalized marginals, while Med-SVGD continues to underestimate target uncertainty. The same conclusion is visible in Figures 13 and 14, where normalized particle quantiles are compared against standard normal quantiles.

Table 1: Coordinatewise marginal variances of the terminal Med-SVGD particle distribution for the d -dimensional examples with $M = 200$, with target variances of π_d included for comparison.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9285							
$d = 2$	0.7921	0.1943						
$d = 3$	0.6803	0.1625	0.0697					
$d = 4$	0.6089	0.1440	0.0593	0.0311				
$d = 5$	0.5532	0.1275	0.0526	0.0271	0.0157			
$d = 6$	0.5190	0.1190	0.0481	0.0243	0.0140	0.0089		
$d = 7$	0.4900	0.1122	0.0449	0.0228	0.0131	0.0081	0.0052	
$d = 8$	0.4753	0.1077	0.0430	0.0215	0.0122	0.0074	0.0047	0.0032

Table 2: Coordinatewise marginal variances of the terminal Ad-SVGD particle distribution for the d -dimensional examples with $M = 200$, with target variances of π_d included for comparison.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9953							
$d = 2$	0.9907	0.2472						
$d = 3$	0.9867	0.2459	0.1095					
$d = 4$	0.9881	0.2467	0.1095	0.0610				
$d = 5$	0.9840	0.2433	0.1096	0.0616	0.0392			
$d = 6$	0.9858	0.2459	0.1090	0.0611	0.0392	0.0269		
$d = 7$	0.9856	0.2463	0.1086	0.0613	0.0390	0.0269	0.0199	
$d = 8$	0.9691	0.2409	0.1085	0.0611	0.0390	0.0268	0.0196	0.0150

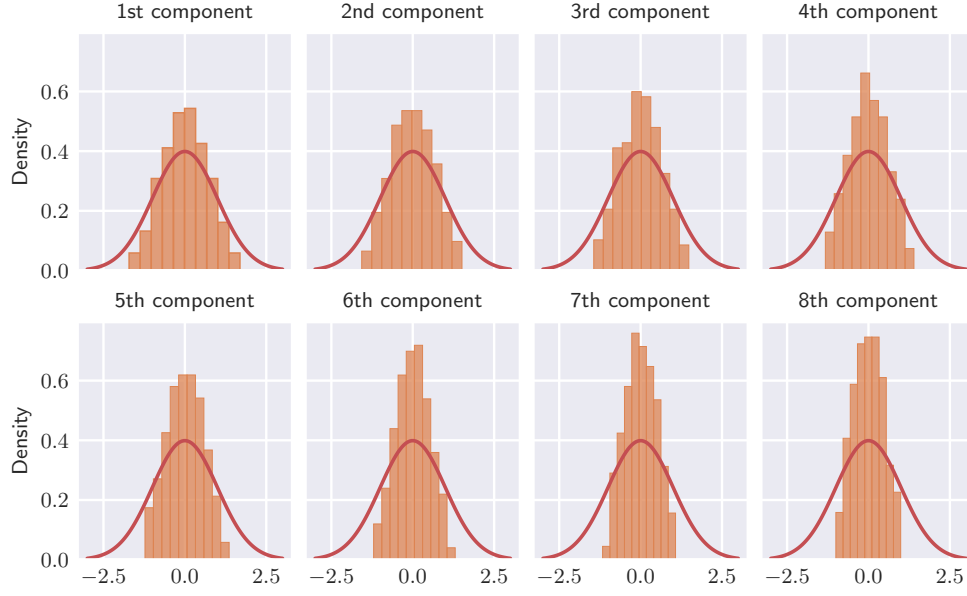


Figure 11: Coordinatewise histograms of the terminal Med-SVGD particles for the eight-dimensional example with $M = 200$, rescaled by target marginal variances; the standard Gaussian density is included for comparison.

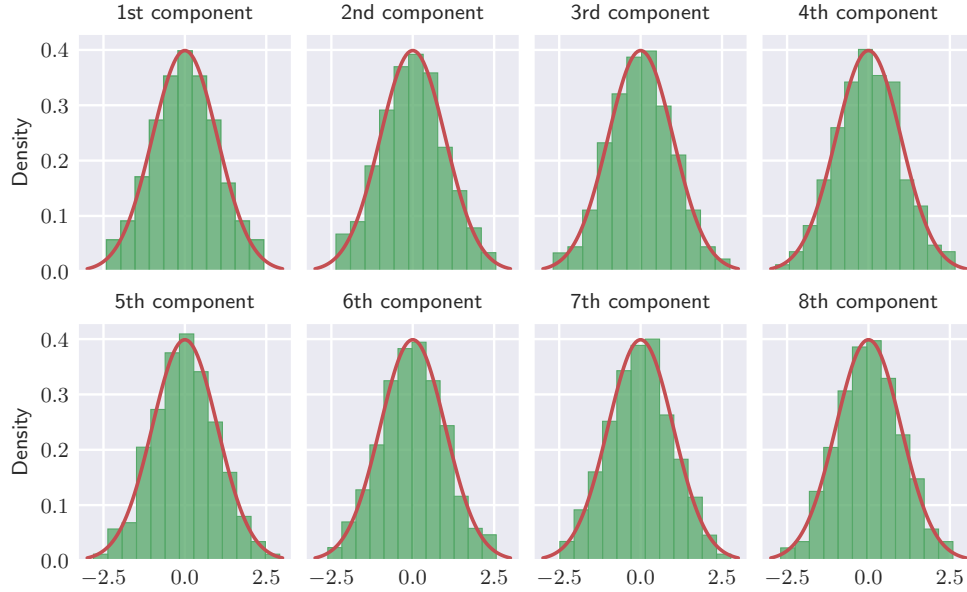


Figure 12: Coordinatewise histograms of the terminal Ad-SVGD particles for the eight-dimensional example with $M = 200$, rescaled by target marginal variances; the standard Gaussian density is included for comparison.

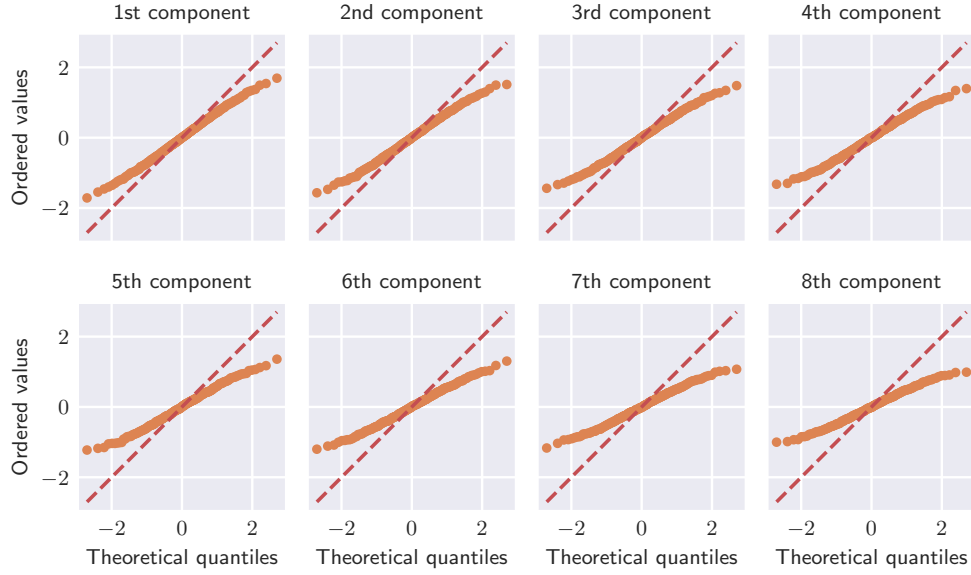


Figure 13: Q-Q plots of the rescaled terminal Med-SVGD marginals for the eight-dimensional example with $M = 200$ against a standard normal distribution; the unit-slope line through the origin is shown for reference.

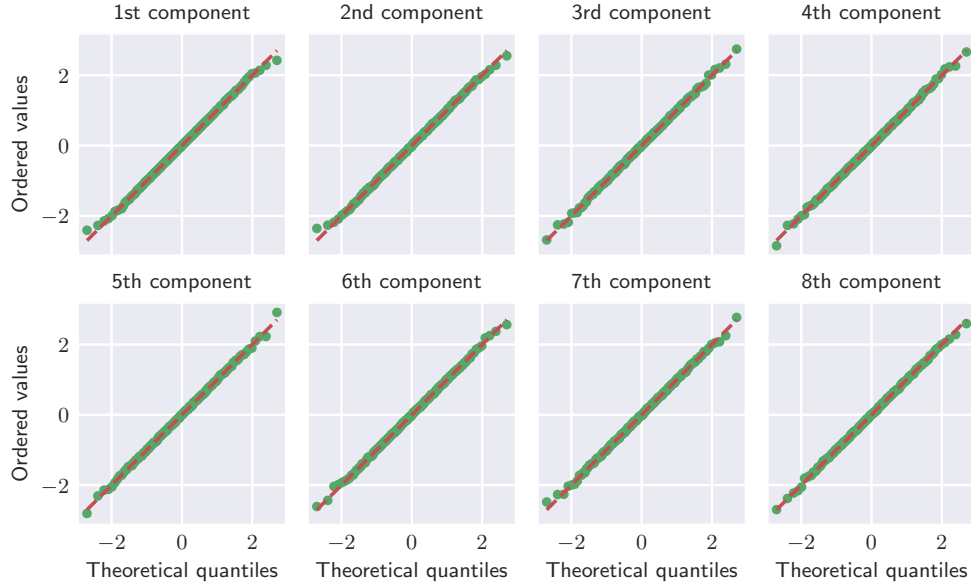


Figure 14: Q-Q plots of the rescaled terminal Ad-SVGD marginals for the eight-dimensional example with $M = 200$ against a standard normal distribution; the unit-slope line through the origin is shown for reference.

A.3 Gaussian process inference

We next evaluate dimensional scaling in the Gaussian-process (GP) inference task proposed in [31]. The GP on $[0, 1]$ is represented by the truncated Karhunen-Loève (KL) expansion $u(s, x) = \sum_{k=1}^{N_x} x_k \psi_k(s)$ with basis functions $\psi_k(s) = \sqrt{2} \sin(k\pi s)$, where $x = (x_1, \dots, x_{N_x})^\top$ and the coefficients are independent with $x_k \sim \mathcal{N}(0, k^{-2})$. The process is observed at N_y equispaced locations in $[0, 1]$, and the task is to infer the coefficients x_k . For fixed N_x and N_y , this gives the linear inverse model $Y = AX + \varepsilon$, where the k -th column of $A \in \mathbb{R}^{N_y \times N_x}$ is $(\psi_k(s_1), \dots, \psi_k(s_{N_y}))^\top$ and $s_i = \frac{i}{N_y}$ for $i = 1, \dots, N_y$. The prior is $X \sim \mathcal{N}(0, \Sigma)$ with diagonal entries k^{-2} , $k = 1, \dots, N_x$, and the noise is independent Gaussian with $\varepsilon \sim \mathcal{N}(0, I_{N_y})$. Reference observations $\bar{y} \in \mathbb{R}^{N_y}$ are generated by drawing $\bar{x} \sim \mathcal{N}(0, \Sigma)$ and setting $\bar{y} = A\bar{x}$.

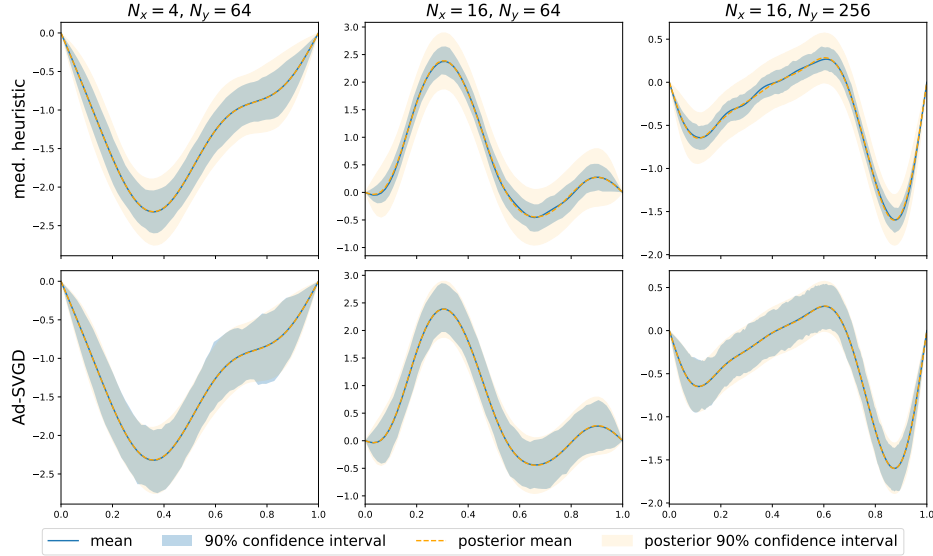


Figure 15: Estimated processes from median-heuristic SVGD and Ad-SVGD with $M = 100$ particles, compared with the posterior for multiple model configurations.

SVGD is then used to approximate the posterior $X | Y = \bar{y}$, comparing the median heuristic against the adaptive bandwidth procedure. Med-SVGD uses kernels $k(x, y) = \exp(-\|x - y\|_1/h)$, whereas Ad-SVGD uses product kernels $k(x, y) = \prod_{i=1}^{N_x} \exp(-|x_i - y_i|/h_i)$ with bandwidth vector $h = (h_1, \dots, h_{N_x})$. Thus Ad-SVGD assigns a distinct bandwidth to each coefficient dimension. Following [18], the cases with $N_x = 16$ use an AdaGrad-type step-size controller.

Figure 15 compares the reconstructed processes with the posterior for $(N_x, N_y) = (4, 64)$, $(16, 64)$, and $(16, 256)$. As the coefficient dimension increases, median-heuristic SVGD underestimates posterior variance, whereas Ad-SVGD provides a closer uncertainty representation. This behavior is stable across the two observation counts considered for the GP. Table 3 quantifies the effect by comparing the trace of the terminal particle covariance with the exact posterior covariance trace. The median heuristic substantially underestimates uncertainty, while Ad-SVGD recovers the covariance trace more accurately.

N_x	4	8	16	16	16
N_y	64	64	64	128	256
theoretical	0.056	0.083	0.086	0.051	0.029
med. heuristic	0.026	0.023	0.022	0.012	0.006
Ad-SVGD	0.055	0.072	0.074	0.044	0.026

Table 3: Trace of the terminal particle covariance ($M = 100$ particles) compared with the theoretical posterior covariance trace for different configurations, averaged over 25 runs.

B Refined theoretical analysis

We can further improve our convergence result under a so-called Stein log-Sobolev inequality, which relates the KL divergence to the KSD. Duncan, Nüsken, and Szpruch [8] introduced this inequality and discussed conditions under which it might hold. Interestingly, in our adaptive setting of kernel selection, we can relax this condition by requiring only one kernel in our entire kernel class to satisfy the log-Sobolev inequality. More precisely, we make the following assumption.

Assumption 5. *Given a kernel class $\{k_\theta \mid \theta \in \Theta\}$, we assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies the generalized Stein log-Sobolev inequality with constant $\lambda > 0$:*

$$2\lambda \text{KL}(\mu \parallel \pi) \leq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu \parallel \pi)$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Using this property, we can quantify convergence in terms of the KL divergence. We emphasize that in this scenario we can give an error bound on $\mathcal{F}(\mu_n)$ even under an approximate condition $\varepsilon_n \leq \bar{\varepsilon}$ without requiring $\varepsilon_n \rightarrow 0$.

Theorem 4. *Suppose that Assumptions 1 to 3 and 5 are satisfied. For any $\alpha > 1$ and $\gamma > 0$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

and

$$\rho := 1 - 2\lambda c_\gamma \in (0, 1) \quad \text{where} \quad c_\gamma := \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2} \right),$$

it holds that

$$\mathcal{F}(\mu_{n+1}) \leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n,$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10) and $\varepsilon_n \geq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \parallel \pi) - \text{KSD}_{\theta_n}^2(\mu_n \parallel \pi)$. In particular, if

1. $\varepsilon_n \leq \bar{\varepsilon}$ for all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{c_\gamma \bar{\varepsilon}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

2. $\varepsilon \leq c_\varepsilon q^n$ for some $q \in (0, 1)$, $c_\varepsilon > 0$ and all $n \in \mathbb{N}$. Under $\rho \neq q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\max\{\rho, q\}^n)$$

and under $\rho = q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\tilde{q}^n)$$

for all $\rho < \tilde{q} < 1$.

3. $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$ for $c_\varepsilon, p > 0$ and all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1 - \rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

Proof. Similar to the proof of Theorem 3, under Assumption 5 and the conditions on α, γ , we have

$$\begin{aligned} \mathcal{F}(\mu_{n+1}) &\leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) + c_\gamma \varepsilon_n \\ &\leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n, \end{aligned}$$

where $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$. Iterating this recursion over $n \in \mathbb{N}$, we deduce the general error bound

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^{n-1-k} \varepsilon_k.$$

We consider the three different cases separately:

1. Under $\varepsilon_n \leq \bar{\varepsilon}$, the first claim follows by the computation of the geometric series

$$\sum_{k=0}^{n-1} \rho^{n-1-k} \leq \sum_{k=0}^{\infty} \rho^k = \frac{1}{1 - \rho}.$$

2. Assuming that $\varepsilon_n \leq c_\varepsilon q^n$ for some $q \in (0, 1)$ and $c_\varepsilon > 0$ we have

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma c_\varepsilon \sum_{k=0}^{n-1} \rho^{n-1-k} q^k.$$

Suppose that $q > \rho$, then we have

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \sum_{k=0}^{n-1} q^{n-1-k} \rho^k = q^{n-1} \sum_{k=0}^{n-1} (\rho/q)^k \leq q^{n-1} \frac{1}{1 - \rho/q} = q^n \frac{1}{q - \rho}$$

where we have used the formula for geometric series. Similarly, in the case of $q < \rho$, we can directly bound

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \rho^{n-1} \sum_{k=0}^{n-1} (q/\rho)^k \leq \rho^{n-1} \frac{1}{1 - q/\rho} = \rho^n \frac{1}{(\rho - q)}.$$

Finally, in the case $q = \rho$, we simply bound $q^k < \tilde{q}^k$ for any $q < \tilde{q} < 1$ and deduce the claim line by line as before.

3. In the setting of $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$, we rewrite the upper bound on $\mathcal{F}(\mu_n)$ as

$$\begin{aligned}
\mathcal{F}(\mu_n) &\leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^k \varepsilon_{n-1-k} \\
&= \rho^n \mathcal{F}(\mu_0) + c_\gamma \left(\sum_{k=0}^{\lfloor n/2 \rfloor} \rho^k \varepsilon_{n-1-k} + \sum_{k=\lfloor n/2 \rfloor + 1}^{n-1} \rho^k \varepsilon_{n-1-k} \right) \\
&\leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1-\rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1-\rho}.
\end{aligned}$$

Here, we have used $\varepsilon_{n-1-k} \leq \frac{c_\varepsilon}{(n-k)^p} \leq \frac{2^p c_\varepsilon}{n^p}$ for all $k \leq \lfloor n/2 \rfloor$ and $\varepsilon_k \leq c_\varepsilon$ for all $k \in \mathbb{N}$.

□